University of Engineering & Technology Peshawar

Computer Programming (C++) Lab

Lab 07

STUDENT NAME	Muhammad Sahal Ayub
REGISTRATION NO:	4042F24
COUSRE:	EE-170LCOMPUTER PROGRAMMING LAB
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LAB TITLE:	ONE DIMENSIONAL ARRAY IN C++

SUBMITTED TO ENGR RIZWAN ULLAH

Learning Objectives:

The objective of this lab is to understand and apply one-dimensional arrays in C++. Students will learn how to:

- Declare and initialize arrays in C++.
- Access and modify individual elements of an array using indices.
- Copy elements from one array to another using loops.
- Perform arithmetic operations (e.g., sum) on array elements.
- Work with character arrays (strings) in C++.

Introduction:

An array is a data structure in C++ that stores a fixed-size sequential collection of elements of the same type. Arrays are stored in contiguous memory locations, where the lowest address corresponds to the first element and the highest address to the last element.

To declare an array:

Syntax:

```
dataType arrayName[arraySize];
```

Example: int age[4]; declares an integer array of 4 elements.

To initialize an array:

```
int age[4] = {23, 34, 64, 75};
```

Array indices start at 0, so age[0] is the first element and age[3] is the last.

Task 1: Array Declaration and Initialization

Declare an integer array called 'numbers' with a size of 5. Initialize it with values 10, 20, 30, 40, 50, and display all elements using a loop.

Code:

```
#include <iostream>
using namespace std;

int main()
{
    // Declare and initialize integer array of size 5
    int numbers[5] = {10, 20, 30, 40, 50};

    // Display array elements using loop
    cout << "Array Elements are:\n";

    for (int i = 0; i < 5; i++)
    {
        // Print each element of array
        cout << "numbers[" << i << "] = " << numbers[i] << endl;
    }

    return 0;
}
```

Output:

```
Array Elements are:
numbers[0] = 10
numbers[1] = 20
numbers[2] = 30
numbers[3] = 40
numbers[4] = 50
```

Task 2: Accessing Array Elements

Declare a character array called 'message' to store a word of your choice. Use a loop to access each character and display it on a separate line.

Code:

```
#include <iostream>
using namespace std;

int main()
{
    // Declare character array to store a word
    char message[] = "HELLO";

    // Display each character on separate line
    cout << "Characters of the word are:\n";

    for (int i = 0; message[i] != '\0'; i++)
    {
        // Print each character until null character
        cout << message[i] << endl;
    }

    return 0;
}
```

Output:

```
Characters of the word are:
H
E
L
L
O
```

Task 3: Copying Arrays

Declare two integer arrays 'source' and 'destination', each of size 5. Initialize 'source' with values, copy elements to 'destination' using a loop, and display both arrays.

Code:

```
#include <iostream>
using namespace std;

int main()
{
    // Declare source and destination arrays
    int source[5] = {5, 10, 15, 20, 25};
    int destination[5];

    // Copy elements from source to destination
    for (int i = 0; i < 5; i++)
    {
        destination[i] = source[i];
    }

    // Display source array
    cout << "Source Array:\n";
    for (int i = 0; i < 5; i++)
    {
        cout << source[i] << " ";
    }

    cout << endl;
```

```
    // Display destination array
    cout << "Destination Array:\n";
    for (int i = 0; i < 5; i++)
    {
        cout << destination[i] << " ";
    }

    return 0;
}
```

Output:

```
Source Array:
5 10 15 20 25
Destination Array:
5 10 15 20 25
```

Task 4: Array Sum

Write a C++ program that prompts the user to enter 5 numbers, stores them in an array, calculates the sum of all numbers, and displays the result.

Code:

```
#include <iostream>
using namespace std;

int main()
{
    // Declare array for 5 numbers
    int numbers[5];
    int sum = 0;

    // Take input from user
    cout << "Enter 5 numbers:\n";

    for (int i = 0; i < 5; i++)
    {
        cin >> numbers[i];

        // Add each number to sum
        sum = sum + numbers[i];
    }

    // Display total sum
    cout << "Sum of all numbers = " << sum << endl;

    return 0;
}
```

Output:

```
Enter 5 numbers:
1
2
3
4
5
Sum of all numbers = 15
```